



Overall system design in embedded value chain

ded design, including development of integrated circuits, FPGAs and SoCs, board design, system design and application engineering.

Software design and FPGA design are India's strengths. India has a relatively small set of people working in the areas of ASIC/SoC and custom design. It has a lot of domain experts who conceptualise embedded systems and work on defining their functionality. If this strength can be leveraged, India can move to the driver's seat in embedded systems.

"The role of domain experts is growing. Hardware today is getting redefined as there is so much more processing power that the constraints of embedded system design in the past, such as memory availability and gate availability in FPGAs, are no longer the concerns, and the intellectual property (IP) that runs on them is the USP of the embedded product. With this significant change in the embedded domain, India has a lead role to play in defining the IPs with its domain expert community and become the driving force of the embedded world," shares Satish Mohan-

ram, business development manager, National Instruments India.

India is doing end-to-end product development right from feasibility study to hardware design, software design, testing and application development. The country has attracted significant business under offshore development centre model. It has designed end-to-end products for almost all multi-billion-dollar companies which lead their respective industry verticals. However, India is not a manufacturing hub yet.

"For India to be in the driver's seat, it has to encourage indigenous product development, and this is possible by creating an eco-system for product development, manufacturing and consumption," says Raghuveer B.K., associate vice president and head-communication products, Infosys.

Software design vis-à-vis hardware design

"By 2015, it is estimated that 70 per cent of the product development effort in embedded systems will focus on the software component—a traditional strength area of the Indian IT-R&D industry. Currently, about 62 per cent of the effort during product development is spent on software-related activities, and this is expected to reach almost 70 per cent over the next six years. The addressable opportunity in embedded software development would therefore grow at a CAGR of 16 per cent going forward, far outpacing the opportunity growth in hardware-related activities," says B. Sridhar, global head-Embedded Systems Group, CMC.

"Some of the key reasons for this trend include strong focus on hardware standardisation and commercial

off-the-shelf (COTS) based development models, which put the onus of differentiation on the software. Need for connectivity, faster product upgrades and advanced user experience are also driving the increasing effort on software in embedded systems," he adds.

Hardware development requires a slightly different approach and involves a higher level of risk. A small error or change can introduce a delay of up to six to eight weeks and certainly add up to the engineering cost. Unlike software development, there are no easy and quick fixes.

"R&D in hardware has been very capital-intensive, unlike software development. With technology shifts happening over the last couple of years, hardware design and development has reached a certain level of maturity. There is more standardisation now and most of the processing has moved from custom silicon to programmable processors or multi-core. This implies that embedded software

India in the value chain of embedded systems

Design

- Low-side design services: Low to medium
- Higher-side product design and development: Very Low

Manufacturing

- Assembly and manufacturing: Low

Testing

- Low

Component

- Electronic components: Very Low
- ICs: Zero
- PCBs: Low

—Ish K. Bhargava, managing director, India Operations, and founder, Spectra Digital Systems



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